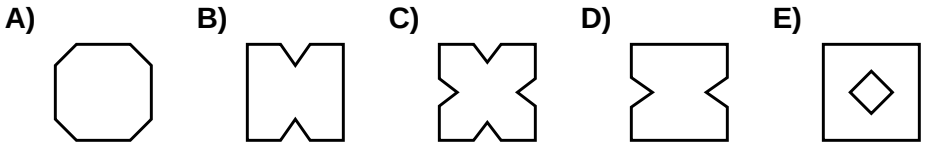
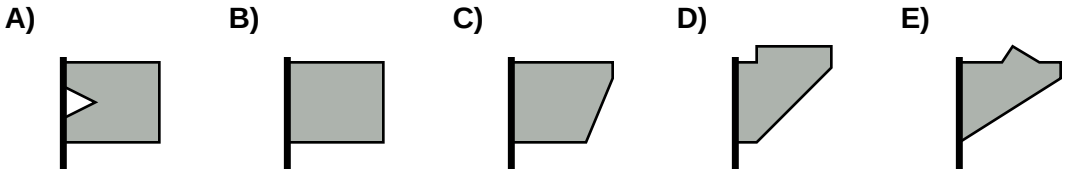
Math Kangaroo Practice Exam - Grades 3-4

20 Randomly Selected Practice Questions (Similar to Contest Problems — Never Originals)

Question 1. A square piece of paper is folded in half vertically and then in half horizontally to form a smaller square. **Exactly one** of the four corners of this folded square is cut off with a straight snip, and the paper is unfolded. Which of the 5 designs below **cannot** be obtained?



Question 2. Maya draws 5 silhouettes of a rectangular flag flying in the wind. One silhouette is impossible unless the flag is ripped along the pole. Which one is it?



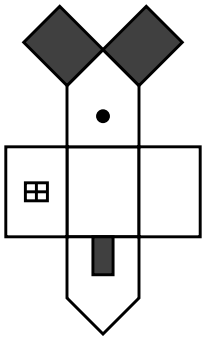
Question 3. The odometer of a bus shows **476358**, where every digit is different. After how many more kilometres will all six digits be different again?

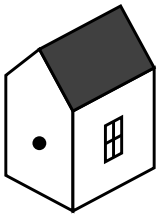
A) 12 **B)** 11 **C)** 9 **D)** 2 **E)** 1

Question 4. All the pages of a puzzle booklet are numbered starting from 1. A total of **63 digits** are used for the page numbers. How many pages does the booklet have?

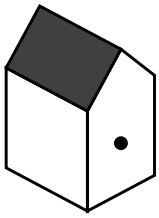
A) 31 **B)** 36 **C)** 38 **D)** 42 **E)** 54

Question 5. This picture is cut out and folded to make a house. Which house does it become? *(The dark parts at the top of the picture are the roof. The shape in the middle is the floor.)*

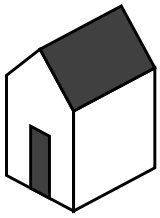


A) 

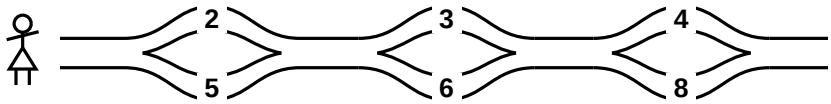
B)

C) 

D)

E) 

Question 6. Sam walked from left to right and **added up** the three numbers he passed along his road. Which of the following could be Sam's total?



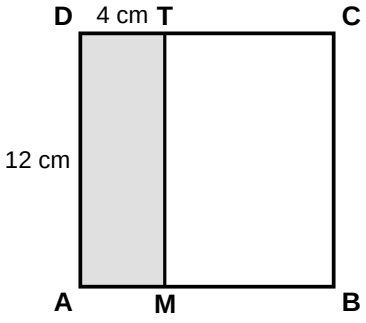
A) 10 **B)** 11 **C)** 14 **D)** 16 **E)** 20

Question 7. Dan has cards 8, 2, and 1; Eva has cards 9, 7, and 8; and Finn has cards 5, 4, and 2. Using +, -, ×, ÷ and each of their own three cards once, who **cannot** obtain **24**?

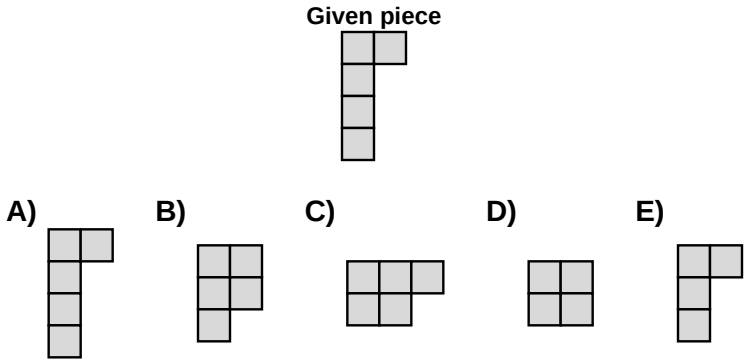
A) Dan **B)** Eva **C)** Finn
D) All can get 24 **E)** Dan and Eva

Question 8. The side of square *ABCD* is 12 cm. The side *AM* of rectangle *AMTD* is 4 cm. How much greater is the perimeter of square *ABCD* than the perimeter of rectangle *AMTD*?

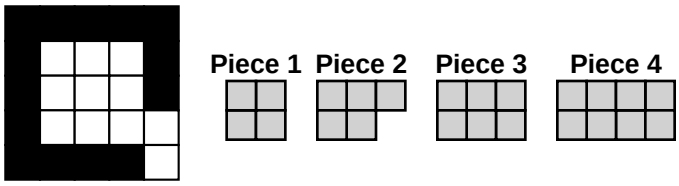
A) 10 cm **B)** 16 cm **C)** 18 cm **D)** 20 cm **E)** 24 cm



Question 9. Which of the shapes (A through E) can be joined together with the given shape to form a solid rectangle without gaps or overlaps?



Question 10. The picture shows a 5 × 5 square with some cells black and the rest white. We want to cover the white area completely without overlaps using two of the numbered pieces (1, 2, 3, 4). Which two pieces should we choose?

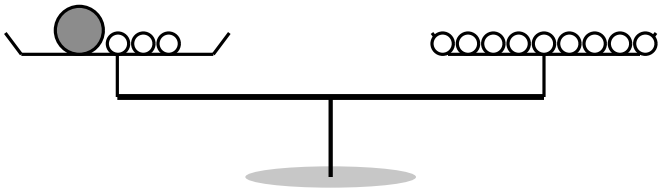


A) 1 and 2 **B)** 1 and 4 **C)** 2 and 4 **D)** 2 and 3 **E)** 3 and 4

Question 11. An athlete threw a javelin 30 dm, 70 cm, and 80 mm on her first try. Her second throw was 114 cm longer than her first throw. How long was her second throw?

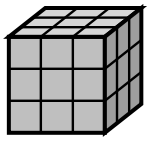
A) 3 m 78 cm **B)** 4 m 84 cm **C)** 4 m 92 cm
D) 5 m 64 cm **E)** 11 m 64 cm

Question 12. The balance is level. On the left plate there is 1 melon (big circle) and 3 oranges; on the right plate there are 9 oranges. How many oranges weigh the same as **2 melons**?



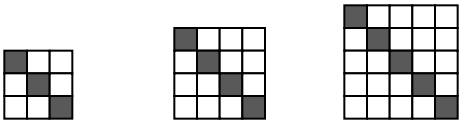
A) 6 **B)** 8 **C)** 9 **D)** 12 **E)** 18

Question 13. A 3 cm × 3 cm × 3 cm wooden cube is painted grey on the outside and cut into 27 small 1 cm cubes. How many of the small cubes have **exactly 1** face painted?



A) 4 **B)** 6 **C)** 8 **D)** 9 **E)** 12

Question 14. The three diagrams show a pattern where the main diagonal is shaded grey (with 6, 12, and 20 white cells). How many white cells will the **next** diagram in the pattern have?



A) 24 **B)** 30 **C)** 32 **D)** 36 **E)** 42

Question 15. Four astronauts board a spacecraft with 4 seats. The Commander must sit in Seat 1 and the Pilot must sit in Seat 2. In how many different ways can the 4 astronauts be seated?

A) 2 **B)** 4 **C)** 6 **D)** 12 **E)** 24

Question 16. Clara built a tunnel 4 cubes deep with a 5 × 5 outer square and a 3 × 3 opening (fig.1). She rearranged the cubes into a tall tower-pyramid (fig.2) consisting of one 5 × 5 base layer, two 3 × 3 middle layers, and three single cubes stacked on top. How many cubes from the tunnel did she *not* use?

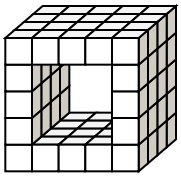


fig.1

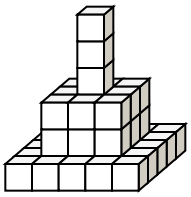
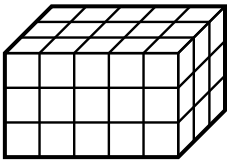


fig.2

A) 12 **B)** 14 **C)** 16 **D)** 18 **E)** 22

Question 17. Anna built this brick (5 cubes wide, 3 cubes tall, 3 cubes deep). The outside is completely red and the cubes inside are blue. How many **red** cubes did she use?



A) 30 **B)** 36 **C)** 39 **D)** 42 **E)** 45

Question 18. Anna, Ben, Clara, and David each own a different pet: a rabbit, a hamster, a turtle, and a canary. Ben and Clara both own furry pets, David owns a bird, and Ben does not have the hamster. Which statement is **not** true?

A) Ben has a rabbit **B)** David has a canary **C)** Clara has a rabbit
D) Anna has a turtle **E)** Clara has a hamster

Question 19. Sophia, who is **younger** than Ethan by two years and one day, was born on May 1, 2014. What is the date of Ethan's birthday?

A) May 2, 2016 **B)** May 2, 2012 **C)** April 30, 2016
D) April 30, 2012 **E)** April 30, 2013

Question 20. A bar code is made of 21 alternating black and white bars, starting and ending with a black bar. Each black bar is either wide or narrow. The number of white bars is 4 more than the number of wide black bars. How many narrow black bars are there?

A) 1 **B)** 2 **C)** 3 **D)** 4 **E)** 5

Answer Key - 20-Question Practice Exam

For generated/2026-10-19-19-03-24.pdf (Randomly sampled from question-pool.txt)

Solutions & kid-friendly explanations

Question 1 (inspired by 2007 #24): C Cutting a single corner of the twice-folded square can cut the 4 outer corners (A), the top and bottom edge midpoints (B), the left and right edge midpoints (D), or the center (E). To get notches on all four edges at once (C), you would have to cut two different corners of the folded square, so C cannot be obtained with one corner cut.

Question 2 (inspired by 2002 #15): A In picture A, there is a triangular gap right along the middle of the flagpole where the flag is supposed to be attached. Since the left edge is fixed to the pole, that gap cannot appear unless the flag is torn.

Question 3 (inspired by 2002 #24): E Adding just 1 km gives 476359, whose digits (4, 7, 6, 3, 5, 9) are already all different! So it happens after just 1 km.

Question 4 (inspired by 2003 #16): B Subtracting the 9 digits used for pages 1–9 leaves 63 - 9 = 54 digits. Those 54 digits form 54 ÷ 2 = 27 two-digit page numbers (pages 10 to 36). So the booklet has 9 + 27 = 36 pages.

Question 5 (inspired by 2003 #12): D Facing the door wall, the window wall is on your right and the plain wall on your left. Only D shows that.

Question 6 (inspired by 2007 #1): D Possible sums are 2+3+4=9, 2+6+4=12, 5+3+4=12, 2+3+8=13, 5+6+4=15, 2+6+8=16, 5+3+8=16, 5+6+8=19. From the choices, only 16 is possible (for example 2 + 6 + 8 = 16).

Question 7 (inspired by 2002 #21): C Dan can get 24 with 8 × (2 + 1) = 24. Eva can get 24 with 9 + 7 + 8 = 24. Finn's cards (5, 4, 2) can make 5 × 4 + 2 = 22 or (5 + 2) × 4 = 28, but never 24. So Finn cannot obtain 24.

Question 8 (inspired by 2002 #6): B Perimeter of square ABCD = 4 × 12 = 48 cm. Perimeter of rectangle AMTD = 2 × (12 + 4) = 32 cm. Difference = 48 - 32 = 16 cm.

Question 9 (inspired by 2007 #8): A The given shape has column heights 4 and 1 (5 squares). Shape A is identical; turned upside-down it has column heights 1 and 4. Joining them gives columns of height 4+1=5 each, creating a 5 × 2 rectangle.

Question 10 (inspired by 2003 #18): D Counting the white cells gives 11 empty cells. The pieces have 4, 5, 6, and 8 squares. Only Piece 2 and Piece 3 have 5 + 6 = 11 squares.

Question 11 (inspired by 2003 #10): C In centimetres, 30 dm = 300 cm, 70 cm = 70 cm, and 80 mm = 8 cm, so the first throw was 300 + 70 + 8 = 378 cm. Adding 114 cm gives 378 + 114 = 492 cm, which is 4 m 92 cm.

Question 12 (inspired by 2002 #9): D Take 3 oranges off each side: 1 melon = 6 oranges, so 2 melons = 12 oranges.

Question 13 (inspired by 2007 #17): B A cube has 6 faces, and in a 3 × 3 × 3 cube only the single center cube on each face has 1 painted face. That gives 6 × 1 = 6 small cubes.

Question 14 (inspired by 2007 #21): B The grids are 3×3 (9 - 3 = 6 white), 4×4 (16 - 4 = 12 white), and 5×5 (25 - 5 = 20 white). The next grid is 6×6 = 36 cells with 6 diagonal grey cells, giving 36 - 6 = 30 white cells.

Question 15 (inspired by 2002 #22): A Seat 1 and Seat 2 are already fixed for the Commander and Pilot. The remaining 2 astronauts can sit in Seats 3 and 4 in 2 × 1 = 2 ways.

Question 16 (inspired by 2002 #20): D The tunnel has 4 layers of 5×5 - 3×3 = 16 cubes, totaling 4 × 16 = 64 cubes. The tower-pyramid uses 25 + (2 × 9) + (3 × 1) = 25 + 18 + 3 = 46 cubes. So 64 - 46 = 18 cubes were not used.

Question 17 (inspired by 2003 #9): D Total = 5 x 3 x 3 = 45. Inside (blue) = 3 x 1 x 1 = 3. Red = 45 - 3 = 42.

Question 18 (inspired by 2002 #18): C David has the bird (canary). The two furry pets are the rabbit and the hamster, belonging to Ben and Clara. Since Ben does not have the hamster, Ben has the rabbit and Clara has the hamster. That leaves the turtle for Anna. Thus, 'Clara has a rabbit' (C) is not true.

Question 19 (inspired by 2007 #12): D Since Sophia is younger than Ethan, Ethan was born 2 years and 1 day BEFORE May 1, 2014. Two years before May 1, 2014 is May 1, 2012, and one day before May 1, 2012 is April 30, 2012.

Question 20 (inspired by 2003 #17): E Out of 21 alternating bars starting and ending with black, there are 11 black bars and 10 white bars. Since 10 white bars is 4 more than the wide black bars, there are 10 - 4 = 6 wide black bars, leaving 11 - 6 = 5 narrow black bars.